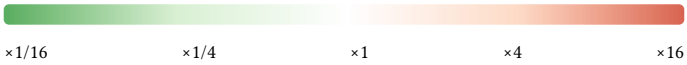
pyAmpliCol at one loop

Batched Born–loop interference · implementation & live benchmarks

2026-10-19 · timings 2026-10-18 · AMD EPYC 9754

One grid, four providers. Each coloured box gives **µs per point** at $B = 1$ or $B = 128$, plus its multiplier against the fastest existing matched reference (ML, Recola or OL). Neutral lines show generation/setup and separately clocked stages at the indicated batch size. Binary64 results include Born, all helicities, full SU(3), UV, model-rule R_2 and all three Laurent coefficients.

Process	pyAmpliCol	MadLoop	Recola	OpenLoops
$u\bar{u} \to e^-e^+$ photon only 1 loop 1 family	$B = 1$ **583.7** **×12.46** vs ML · 46.86 µs $B = 128$ **147.4** **×3.17** vs ML · 46.49 µs **Generation 2.43 s** source + complete plan first $B = 3$: 0.022 s peak RSS 132 MiB **Stages at $B = 32$ · µs/point** Numerator 24.67 NinjaRS only 0.92 Scalar masters 0.35 Other work 71.69 Clocked total 96.84	$B = 1$ **46.86** **×1.00** vs ML · 46.86 µs $B = 128$ **46.49** **×1.00** vs ML · 46.49 µs **Generation 198.14 s** export + static compilation includes bundled reducers **Stages at $B = 32$ · µs/point** Numerator 6.47 Reduction + masters 35.86 Shared HELAS is outside the numerator clock.	$B = 1$ **1878** **×40.07** vs ML · 46.86 µs $B = 128$ **1784** **×38.37** vs ML · 46.49 µs **Generation 0.135 s** parameters + define + generate	$B = 1$ — Timing not recorded $B = 128$ — Timing not recorded **Setup —** matched control unavailable
$u\bar{u} \to d\bar{d}$ 14 loops 8 families	$B = 1$ **1733** **×6.90** vs ML · 251.3 µs $B = 128$ **764.0** **×3.00** vs ML · 254.3 µs **Generation 7.09 s** source + complete plan first $B = 3$: 0.063 s peak RSS 180 MiB **Stages at $B = 32$ · µs/point** Numerator 294.0 NinjaRS only 7.76 Scalar masters 9.05 Other work 381.0 Clocked total 691.4	$B = 1$ **251.3** **×1.00** vs ML · 251.3 µs $B = 128$ **254.3** **×1.00** vs ML · 254.3 µs **Generation 14.43 s** export + static compilation **Stages at $B = 32$ · µs/point** Numerator 55.76 Reduction + masters 197.1 Shared HELAS is outside the numerator clock.	$B = 1$ **2651** **×10.55** vs ML · 251.3 µs $B = 128$ **2672** **×10.51** vs ML · 254.3 µs **Generation 0.253 s** parameters + define + generate	$B = 1$ **547.7** **×2.18** vs ML · 251.3 µs $B = 128$ **563.3** **×2.22** vs ML · 254.3 µs **Setup 0.038 s** prebuilt library · register + start source generation not timed
$u\bar{u} \to t\bar{t}$ 14 loops 9 families	$B = 1$ **2006** **×5.70** vs ML · 352.0 µs $B = 128$ **745.0** **×1.98** vs ML · 376.2 µs **Generation 6.39 s** source + complete plan first $B = 3$: 0.091 s peak RSS 183 MiB **Stages at $B = 32$ · µs/point** Numerator 334.9 NinjaRS only 16.10 Scalar masters 18.28 Other work 520.8 Clocked total 889.8	$B = 1$ **352.0** **×1.00** vs ML · 352.0 µs $B = 128$ **376.2** **×1.00** vs ML · 376.2 µs **Generation 17.05 s** export + static compilation **Stages at $B = 32$ · µs/point** Numerator 94.48 Reduction + masters 256.9 Shared HELAS is outside the numerator clock.	$B = 1$ **2908** **×8.26** vs ML · 352.0 µs $B = 128$ **3053** **×8.12** vs ML · 376.2 µs **Generation 1.122 s** parameters + define + generate	$B = 1$ **654.6** **×1.86** vs ML · 352.0 µs $B = 128$ **686.4** **×1.82** vs ML · 376.2 µs **Setup 0.010 s** prebuilt library · register + start source generation not timed



Logarithmic scale · lower is faster · white is parity · red is slower

MEASUREMENTS · SAME GRID, CONTINUED

Complete evaluation · remaining processes

Warm boxes: µs per point at $B = 1$ and $B = 128$. Neutral preparation: seconds. Instrumented stage lines: $B = 32$ for two-to-two, $B = 128$ for five-leg; no stage speed ratios.

Process	pyAmpliCol	MadLoop	Recola	OpenLoops
$u\bar{u} \to gg$ 45 loops 20 families	$B = 1$ **6835** **×14.77** vs ML · 462.8 µs $B = 128$ **3100** **×8.34** vs ML · 371.5 µs **Generation 21.16 s** source + complete plan first $B = 3$: 0.616 s peak RSS 321 MiB **Stages at $B = 32$ · µs/point** Numerator 1778 NinjaRS only 21.52 Scalar masters 17.65 Other work 1389 Clocked total 3196	$B = 1$ **462.8** **×1.00** vs ML · 462.8 µs $B = 128$ **371.5** **×1.00** vs ML · 371.5 µs **Generation 29.11 s** export + static compilation **Stages at $B = 32$ · µs/point** Numerator 205.5 Reduction + masters 127.3 Shared HELAS is outside the numerator clock.	$B = 1$ **5343** **×11.55** vs ML · 462.8 µs $B = 128$ **4470** **×12.03** vs ML · 371.5 µs **Generation 0.183 s** parameters + define + generate	$B = 1$ **1702** **×3.68** vs ML · 462.8 µs $B = 128$ **1696** **×4.56** vs ML · 371.5 µs **Setup 0.015 s** prebuilt library · register + start source generation not timed
$gg \to gg$ 186 loops 52 families	$B = 1$ **33 656** **×12.98** vs ML · 2593 µs $B = 128$ **14 280** **×6.89** vs ML · 2073 µs **Generation 98.74 s** source + complete plan first $B = 3$: 11.532 s peak RSS 1085 MiB **Stages at $B = 32$ · µs/point** Numerator 9185 NinjaRS only 88.42 Scalar masters 68.76 Other work 4515 Clocked total 14 167	$B = 1$ **2593** **×1.00** vs ML · 2593 µs $B = 128$ **2073** **×1.00** vs ML · 2073 µs **Generation 30.14 s** export + static compilation **Stages at $B = 32$ · µs/point** Numerator 1916 Reduction + masters 824.8 Shared HELAS is outside the numerator clock.	$B = 1$ **36 056** **×13.91** vs ML · 2593 µs $B = 128$ **33 245** **×16.03** vs ML · 2073 µs **Generation 0.170 s** parameters + define + generate	$B = 1$ **11 875** **×4.58** vs ML · 2593 µs $B = 128$ **16 233** **×7.83** vs ML · 2073 µs **Setup 0.013 s** prebuilt library · register + start source generation not timed
$u\bar{u} \to ggg$ 556 loops 119 families	$B = 1$ **194 468** **×5.39** vs ML · 36 056 µs $B = 128$ **125 357** **×3.74** vs ML · 33 494 µs **Generation 862.87 s** source + complete plan first $B = 3$: 18.424 s peak RSS 5329 MiB **Stages at $B = 128$ · µs/point** Numerator 55 494 NinjaRS only 268.1 Scalar masters 288.4 Other work 73 757 Clocked total 129 626	$B = 1$ **36 056** **×1.00** vs ML · 36 056 µs $B = 128$ **33 494** **×1.00** vs ML · 33 494 µs **Generation 23.89 s** export + static compilation **Stages at $B = 128$ · µs/point** Numerator 26 380 Reduction + masters 2466 Shared HELAS is outside the numerator clock.	$B = 1$ **81 544** **×2.26** vs ML · 36 056 µs $B = 128$ **88 558** **×2.64** vs ML · 33 494 µs **Generation 0.724 s** parameters + define + generate	$B = 1$ **36 928** **×1.02** vs ML · 36 056 µs $B = 128$ **43 380** **×1.30** vs ML · 33 494 µs **Setup 0.011 s** prebuilt library · register + start source generation not timed

Original five processes retain revision fa599885 the five-leg row uses f89249d2. These are complete model-rule development observations; see the continued grid and measurement boundaries on the next page.

MEASUREMENTS · SAME GRID, CONTINUED

Real-mass electroweak control

Full Gmu electroweak correction, real masses, source compiler revision 85fb0900. Warm boxes: µs per point. Generation: seconds. Separate stage clocks: $B = 128$.

Process	pyAmpliCol	MadLoop	Recola	OpenLoops
$e^-e^+ \to \mu^-\mu^+$ full EW · Gmu 98 loops 24 families	$B = 1$ **6589** **×2.12** vs Recola · 3105 µs $B = 128$ **2922** **×0.99** vs Recola · 2946 µs **Generation 115.82 s** source + complete plan first $B = 3$: 0.179 s peak RSS 711 MiB **Stages at $B = 128$ · µs/point** Numerator 1539 NinjaRS only 31.27 Scalar masters 67.02 Other work 1284 Clocked total 2926	$B = 1$ — Timing not recorded $B = 128$ — Timing not recorded **Generation —** not yet recorded **Stages at $B = 128$ · µs/point** Numerator — Reduction + masters — Shared HELAS is outside the numerator clock.	$B = 1$ **3105** **×1.00** vs Recola · 3105 µs $B = 128$ **2946** **×1.00** vs Recola · 2946 µs **Generation 0.175 s** parameters + define + generate	$B = 1$ — Timing not recorded $B = 128$ — Timing not recorded **Setup —** matched control unavailable

The matched Recola setup retains actual poles and the same Gmu inputs. MadLoop validates complete components; its timing and OpenLoops timing are not recorded here. No provider speed is inferred from correctness tests.

The EW row uses $G_\mu = 1.16639 \times 10^{-5}$ GeV^{-2}, $m_W = 80.419$, $m_Z = 91.188$, $m_H = 125$, $m_t = 173.3$ GeV, zero widths and light masses, and $\mu_R = 1000$ GeV. Its full source counterterms are retained.

Numerator, reduction and masters

pyAmpliCol numerator includes loop DAGs, callbacks and coefficient-head preparation. NinjaRS excludes callbacks and masters; other work includes states, Born, UV/R_2, exact kinematics, assembly and transport. ML excludes shared HELAS from numerator and includes interfaces/retries in reduction + masters.

Separate clocks, separate observations

Warm totals use uninstrumented calls. Stage medians come from separate clocked calls and need not sum to the median clocked total. Master clocks retain the adapter cache policy. Rust dev: local opt3 / dependencies opt2, checks on, no LTO; Community opt2. Raw timing samples are retained.

Five warm calls on repeated three-point streams; no cached event results in pyAmpliCol. CPU 382, one thread/shared host. pyAmpliCol includes decimal ingress/ Python scheduling with rescue off; providers retain native-array input and normal rescue/cache policies. Formatting is excluded. Preparation: CPU 380, ML DY 381, Recola DY and EW 382; no cache flush or uncertainty estimate. Recola photon-only DY passes analytic checks; OL DY is absent.

Development comparison: the measured QCD rows remain slower than ML. Acceptance requires distinct points, matched boundaries/stability, faster than ML and within 1.30 × Recola/OL per designated high-multiplicity case; generation has no gate. OL actual-pole mode runs the amplitude three times with Collier cache off. Original loop kernels use `-O0`; a separate `-O3` gg trial showed no clear win.

MEASUREMENTS · MATCHED LEADING COLOUR

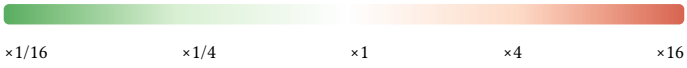
Leading colour and its full-colour counterpart

An internal comparison. Each warm cell pairs LC with a new full-colour measurement of the same process and R_2 strategy. Its colour and multiplier refer only to that full-colour run; no external provider enters this page. Both binary64 paths return Born and all three completed Laurent coefficients.

Process / strategy	Single point · $B = 1$	Native batch · $B = 128$	Generation · seconds
$u\bar{u} \to e^- e^+$ photon only Model-rule R_2	LEADING COLOUR 359.6 µs/point Full: **351.2** µs/point LC / full: ×**1.02**	LEADING COLOUR 103.4 µs/point Full: **98.13** µs/point LC / full: ×**1.05**	LC: **1.61** s Full: **1.61** s First $B = 3$, LC / full: 0.009 / 0.010 s
$u\bar{u} \to d\bar{d}$ Model-rule R_2	LEADING COLOUR 1635 µs/point Full: **1536** µs/point LC / full: ×**1.06**	LEADING COLOUR 609.0 µs/point Full: **646.7** µs/point LC / full: ×**0.94**	LC: **6.05** s Full: **5.27** s First $B = 3$, LC / full: 0.058 / 0.057 s

Stages · LC / full · µs/point

	$u\bar{u} \to e^- e^+$	$u\bar{u} \to d\bar{d}$
Numerator	25.94 / 23.11	287.4 / 303.1
NinjaRS only	0.66 / 0.64	6.30 / 6.17
Masters	0.20 / 0.21	6.83 / 6.53
Other	76.81 / 70.79	355.4 / 368.5
Clocked total	103.8 / 94.67	658.3 / 693.9



Logarithmic scale · lower is faster · white is parity · red is slower

Independent fresh process for each colour mode, revision 95c4ed50, CPU 379, one thread/shared host. Five timed calls use the repeated three-point controls; decimal ingress and Python batch scheduling are included, checks and output formatting excluded. Generation includes source capture and the complete plan; first evaluation is separate. These are development observations, not an LC performance target or a replacement for the full-colour snapshot on pages 1–3.

Fixed-N_f LC keeps the common highest formal-N_c power before setting $N_c = 3$. Full and LC retain the current graph schedules; this is not yet numerator pruning. Quark controls use the fixed-SU(3) full path and formal LC path. The gluon control uses the certified direct strategy and formal algebra for both paths, avoiding an unsupported continuation of the model's four-gluon R_2 coefficients. Full components pass independent references; LC checks use the certified Born, pole and UV/field relations, with complete component reconstruction.

MEASUREMENTS · SAME COLOUR GRID, CONTINUED

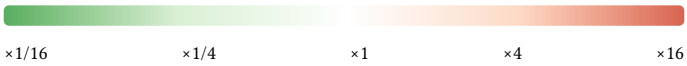
Leading colour · massive quarks and gluons

Warm cells: µs per point. Neutral B128 stage pairs: LC / full, from separate instrumented calls; medians need not add. Generation and first evaluation: seconds.

Process / strategy	Single point · $B = 1$	Native batch · $B = 128$	Generation · seconds
$u\bar{u} \to t\bar{t}$ Model-rule R_2	LEADING COLOUR **1729** µs/point Full: **1941** µs/point LC / full: ×**0.89**	LEADING COLOUR **723.7** µs/point Full: **730.0** µs/point LC / full: ×**0.99**	LC: **5.64** s Full: **5.25** s First $B = 3$, LC / full: 0.078 / 0.084 s
$gg \to gg$ Direct R_2	LEADING COLOUR **34 688** µs/point Full: **34 287** µs/point LC / full: ×**1.01**	LEADING COLOUR **14 570** µs/point Full: **18 843** µs/point LC / full: ×**0.77**	LC: **126.59** s Full: **161.07** s First $B = 3$, LC / full: 13.941 / 13.612 s

Stages · LC / full · µs/point

	$u\bar{u} \to t\bar{t}$	$gg \to gg$
Numerator	291.5 / 293.5	12 005 / 11 703
NinjaRS only	10.00 / 10.01	161.2 / 144.0
Masters	13.06 / 12.46	110.4 / 104.3
Other	433.1 / 437.9	2895 / 4050
Clocked total	747.9 / 752.4	15 302 / 16 133



Logarithmic scale · lower is faster · white is parity · red is slower

What is and is not measured. Every timed call computes the complete selected observable. LC changes the retained formal-colour powers; current graph schedules still run. These same-strategy internal pairs do not compare direct and model R_2, nor replace the model-rule provider grid. Stage scopes match that grid: numerator includes DAGs and reducer callbacks/head preparation; NinjaRS excludes callbacks and scalar masters. Other work includes states, Born weighting, counterterms, kinematics, assembly and instrumentation.

Opt-in projector JIT. Saved family bilinears now bind through the existing tiled coefficient plan when `projector_engine="jit"` is requested in binary64. The interpreter remains the default and the DoubleFloat/AP path. A fresh complete gluon check at `d6c05e08`, on 128 distinct exact energy-scaled control points, gave $B = 128$ projection medians of 1.75 to 0.52 ms/point and complete calls of 13.28 to 11.85 ms/point. Five paired CPU379 blocks retain visible variation; independent physics gates passed unchanged. Saved ordering, retained/portable bindings, tail/empty batches and precision checks pass. JIT is not bit-exact; earlier output-wise DD evidence remains available. No algebra, arena or factorization change is included. These observations do not replace the timing tables.

ARCHITECTURE · PHYSICAL CONTRACT

The quantity being computed

The new engine targets one-loop amplitudes interfered with a Born amplitude. It does not compute the squared one-loop amplitude and does not admit loop-induced production. A Born zero at an individual point or helicity remains a valid event of a supported process.

For Born order a, loop order b and squared coupling order $s = a + b$, the ordinary uncorrelated observable is

$$V_{s,k} = \mathcal{N} \sum_{a+b=s} 2\Re\left[B_a^\dagger G_{BL} L_{b,k}\right], \quad k \in \{0, -1, -2\}.$$

The rectangular matrix G_{BL} connects the actual Born and loop colour bases. The normalization $\mathcal{N}$ includes declared initial averages and final identical-particle factors. Couplings are included; the result is neither divided by the Born value nor normalized by an extracted strong coupling.

The returned coefficient order is **(finite, single pole, double pole)** in the MadLoop convention,

$$V = C_\varepsilon \left(V_0 + \frac{V_{-1}}{\varepsilon} + \frac{V_{-2}}{\varepsilon^2} \right), \quad C_\varepsilon = \frac{(4\pi)^\varepsilon}{\Gamma(1-\varepsilon)}.$$

The finite conversion used in the matched Recola controls is $V_0^{\mathrm{ML}} = V_0^{\mathrm{Recola}} + \left(\frac{\pi^2}{6}\right) V_{-2}$; both poles are unchanged. Such conversions act on the Laurent series, not on a process-dependent fitted factor. Scheme, scale and finite renormalization conventions are retained separately.

Coupling selection is a graph property

`loop_coupling_orders` requests perturbative corrections. Eligibility depends on the actual loop interactions and internal species, followed by explicit Born–loop pairings. An order difference alone cannot identify the correct graphs. In the mixed four-quark control,

$$B = B_{20} + B_{02}, \quad L = L_{40} + L_{22} + L_{04},$$

the squared order $(4, 2)$ receives both $B_{20}^* L_{22}$ and $B_{02}^* L_{40}$. Their identities survive through counterterm matching and output assembly. Joint QCD/QED selection also admits mixed graphs absent from either singleton selection. Independent MadLoop graph controls give 16 QCD, 15 QED and 35 joint loops for the retained mixed model.

What the measured processes include

Control	Initial / identical normalization	Physical completion
$u\bar{u} \to e^- e^+$	$\frac{1}{36}$	Photon Born; QCD virtual
$u\bar{u} \to d\bar{d}$	$\frac{1}{36}$	Local UV and R_2
$u\bar{u} \to t\bar{t}$	$\frac{1}{36}$	Local and external-field UV; R_2
$u\bar{u} \to gg$	$\frac{1}{72}$	Local/field UV; R_2
$gg \to gg$	$\frac{1}{512}$	Local/field UV; R_2
$u\bar{u} \to ggg$	$\frac{1}{216}$	Local/field UV; R_2
$e^- e^+ \to \mu^- \mu^+$	$\frac{1}{4}$	Full EW; real-mass Gmu; local UV/ R_2

The QCD scattering controls use $\alpha_s = 0.118$, $\mu_R = 1000$ GeV, stable $m_t = 173$ GeV, five renormalization flavours and six quarks in loops; the other quark masses vanish under explicit control restrictions. Drell–Yan uses $\alpha^{-1} = 132.507$. Initial spin and colour are averaged, and physical external helicities are summed. Two final gluons include $\frac{1}{2}$, three include $\frac{1}{6}$.

ARCHITECTURE · MODEL AND COUNTERTERMS

Preserve the model before lowering it

Loop models contain information that a tree-only loader can discard: Laurent-valued parameters and couplings, UV and R_2 categories, loop-particle groups, external-field factors and duplicate vertex names. The upstream loader's raw source model retains these fields and positional identities in ordinary model JSON. pyAmpliCol consumes it through a thin adapter, with one stored source copy.

Raw source
Exact declarations
Species and loop groups

→

Prepared model
Scalar programs
Colour × Lorentz entries

→

Process plan
Admitted graphs
Matched UV / R_2 / fields

Exactness begins at input

Literal decimal and Laurent source text, exact overrides and fixed parameters retain their scalar owner. Imported results of arbitrary Python computation or mutation do not acquire exactness. Symbolica performs substitution, series and expression preparation. Source helpers retain their branches; native evaluation checks each value and active diagnostic. A nonfinite or unsupported branch cannot silently succeed. Per-point scales reach the same counterterm scalar program.

Lowering omits a basis only when exact coupling bounds prove it irrelevant; source, restrictions and species eligibility remain retained. The explicit `loop.model` contract declares joint correction coverage, UV/R_2 completeness, dimensional regularization, external states, gamma-five, renormalization, masses and finite restoration. It remains `declared-not-certified`: model names and counterterm presence do not prove completeness. Public generation is still being connected.

Tensor structure and physical frame

Each basis entry retains its scalar coefficient, native Lorentz tensor, native colour tensor and ordered particle ports. Multi-colour quartic vertices remain sums of local terms. There is no reconstruction of tensor algebra in pyAmpliCol: Idenso/Spenso/FeynKit own contractions, traces, component libraries and index transformations.

Source spinor slots require an explicit map to native wavefunctions. The FFV map renames typed external Dirac ports simultaneously; the FFS map also exchanges `ProjP`/`ProjM`, with 580 independent ALOHA controls. Neither rewrites momentum labels or supplies a universal gamma-five transpose. Unvalidated structures remain unsupported.

Counterterms follow admitted loops

UV-tree terms, local loop counterterms and external-field factors have different eligibility rules. A standalone counterterm tree is insufficient evidence that its loop correction belongs to the request. Local R_2/UV matching retains the actual attachment currents, ordered binding legs, local coupling order and allowed loop-species alternatives. Wildcards and repeated alternatives preserve their source semantics and multiplicity. Fresh native tree graphs provide the insertion signs and propagators. Ordinary Bose slot classes are identified only after exact native tests of the complete vertex under generating transpositions. This repairs 39 missed insertions in the five-leg $u\bar{u} \to ggg$ control. Complete Born, bare, UV and R_2 components now agree with MadLoop at the reference accuracy in double, DoubleFloat and AP80; exact parameter-update controls also pass.

External-field factors multiply the applicable Born currents with each physical leg retained. Local and field UV remain separately inspectable, then join the same ordered interference. Model-rule R_2 currents already contain their loop normalization; the assembly does not multiply them by a second loop measure.

The bounded Gmu continuation offers an explicit `ufo-standard` logarithm convention. The default preserves the source helper; either choice retains the original model and does not certify complete CMS physics.

Loader 1.0.1 is published. Lossless source import, JSON and finite tree projection pass QCD/EW controls. `sqrt (2)` and real/imaginary parsing are repaired; SM assets need no changes. pyAmpliCol uses the released source API.

ARCHITECTURE · RECURRENCE AND FAMILIES

Local tensors, reusable coefficient currents

The production direction is a dedicated loop recurrence. FeynKit supplies the graphs, directed incidence, loop signs, symmetry factors, cuts and affine momentum routing. pyAmpliCol turns those local ingredients into a coefficient dataflow graph and schedules its numerical evaluation.

A current retains the coefficients of a symmetric polynomial in the physical loop momentum. For a local tensor T and previous current J, the next step is the contraction $T \cdot J$, followed by coefficient extraction using Symbolica. Only the local transition is prepared symbolically. A whole-diagram symbolic trace is used in small independent oracles, not as the numerical recurrence.

Coefficient dataflow. External states and tree attachments feed local transitions; loop closure produces ordered coefficient outputs. Native saved evaluator instructions update these outputs in batches. Each original graph retains its colour terms, coupling sector and routing map even when it shares an identical numerical schedule with another graph.

Topology remains attached to the numerator

Reduction needs a denominator set as well as a polynomial. Each graph carries its routed propagators and a dependency-proved map into an integral family. Numerators are transported through that map before combination. Cyclic or reversed order is not inferred from a loose list of masses: momenta, incidence, external attachments and the actual transformation must agree.

The family plan retains bridges and local tree denominators. Native graph factors carry fermion and ghost signs. Multi-term local vertices are expanded into ordered recurrence outputs, while the original graph/family identities remain intact. This separates a numerical reuse opportunity from a physical multiplicity or symmetry factor.

Contract before reducing when linearity permits

For an interference observable, Born amplitudes and the rectangular colour weights can multiply numerator coefficients before integration. Compatible graphs, helicities and colour terms then share a reduction within their family and coupling sector. The reducer sees the coefficient combination needed by the observable, rather than every independent loop amplitude.

Complex weights are retained until the correctly directed bilinear is formed. Taking a real part too early loses information needed by colour histories and other off-diagonal observables. Counterterm interference uses the same Born currents and colour conventions, with its own local scalar factors.

Sharing that is safe under parameter updates

Compiled kernel reuse keys the saved instructions, settings, carrier precision and output layout. Evaluators may share immutable compiled code while owning independent work buffers. Whole-schedule numerical reuse additionally requires identical ordered inputs under an exact model-binding proof. Equal floating values at one point are insufficient.

The current process cache shares compatible Born, loop and counterterm kernels; exact whole-schedule groups cover Born and loop execution. It does not cache event results between calls. Complete unprojected model-rule QCD plans now retain exact instructions and bindings across arithmetic views. Rebinding creates fresh evaluators, reducers and buffers without repeating graph, colour or coefficient preparation. Colour-weight changes invalidate the cached plan. The bounded full-colour direct-R_2 gluon plan also rebinds its retained dimensional DAG and physical/evanescent sectors, including UV and field completion. Source-owned complete QCD views also retain their exact scalar definitions and parameter snapshots across carrier rebinding. Formal-colour and correlated plan reuse remain outside this checkpoint; reducer Laurent preparation remains a cold binding cost.

Separate sharing experiment. On distinct exact energy rescalings of gluon controls, five alternating calls at $B = 32, 128$ gave complete medians 15.03 ⟶ 19.97 / 15.55 ⟶ 21.91 ms per point. Exact interning reduced 1,963 nodes to 1,358 with bit-identical outputs, but native execution slowed. A cache-locality explanation remains an inference; global interning was not adopted. CPU 379/shared host; separate from headline data.

Graph-local arena experiment. Existing lifetime allocation reduced retained arena capacity by 48.4% without changing instructions or 16-row tiles. Complete $B = 32, 128$ medians were 14.23 ⟶ 13.55 / 13.60 ⟶ 14.06 ms per point. Coefficients and complete components were bit-identical; DD/AP80 layout checks passed. No speed benefit was established; the production layout is unchanged. Both plans coexisted, capacity excludes dependency buffers, and raw DAG clocks remain available. Same distinct-point protocol and CPU.

ARCHITECTURE · COLOUR AND OBSERVABLES

Full colour without dense event tensors

The current complete QCD path contracts colour at plan construction using native fixed-SU(3) tensors. It forms the properly scoped bra–ket expression, uses dependency colour simplification and caches the exact pair. The numerical event path receives the resulting rectangular weights. Dense SU(3) components remain an independent small-control oracle, not the production colour sum.

Closed residual contractions use the dependency's existing projector expansion, trace/chain operations and canonicalization. Minimal upstream fixes address specific demonstrated tensor-lowering or invariant-contraction defects. No private group identities or numerical colour fitting are inserted into the application.

LC and NLC are structural projections

For a certified formal-N_c model, all contributions belonging to one squared order and operator family are combined before finding the common highest power P. With N_f held fixed,

$$\mathrm{LC} : N_c^P, \quad \mathrm{NLC} : N_c^P, N_c^{P-1}, N_c^{P-2}.$$

Odd suppression powers are retained. The same P spans the relevant helicities, epsilon sectors and source contributions; choosing a different leading power for each term would change the approximation. Exact native algebra handles Laurent coefficients and semantic C_A, C_F, T_R relations.

Complete bounded quark-process colour projections have controls. The active `loop_sm` four-gluon R_2 model contains fixed-$N_c = 3$ coefficients. Its full-SU(3) validity does not imply a formal continuation, and numerical constants are never promoted heuristically to N_c. Validated direct completion can provide a different continuation only within its certified domain.

Directed colour and spin sources

Correlated virtual observables require both rectangular cross terms. For operator histories A and B, their generic form is

$$B_A^\dagger G L_B + L_A^\dagger G B_B.$$

For unequal histories this is not generally $2\mathfrak{R}\left(B_A^\dagger G L_B\right)$; the result can be complex. Existing colour-connection metadata and native charge tensors provide the operator action and signs. Current internal controls compose bounded histories with Born, bare loop, local counterterms and field factors.

Literal complex spin-source vectors enter the admitted physical external ports without an implicit conjugation, normalization or projection. Spectator states are summed and a replaced leg is counted once. The source/current frame and crossing factors remain explicit. Scalar and massive-vector layouts reuse native wavefunctions; a generic source API is not yet certified by these controls.

Flavour mapping and FFT remain distinct work

An exact species-symmetry certificate now compares the entire retained ordinary, UV, R_2 and external-field plan, including prepared native structures and parameter bindings. Independent massless d and s catalogues pass; unequal masses, Yukawas or UV logarithms reject. This is a prerequisite for mapping, not yet external-leg, momentum, spin-source or colour-history transport.

Optional FFT contraction is an agreed feature but is not part of the measured path. Its cost model, exact observable reconstruction and composition with the other features remain acceptance gates.

ARCHITECTURE · NATIVE BATCHES AND PRECISION

One numerical path, three real carriers

The internal numerical core uses binary64, DoubleFloat and arbitrary-precision carriers throughout coefficient evaluation, kinematics, NinjaRS reduction, scalar masters, counterterms and contractions. It does not compute a binary64 answer and promote it afterward. Fixed-carrier precision metadata follows the actual numerical types; arbitrary precision carries its requested precision.

Exact decimal or rational input is parsed before carrier conversion. S-matrix and admitted on-shell invariants are constructed through the exact input boundary; source mass/coupling formulas are evaluated natively in the chosen carrier. Controls with differences below binary64 resolution verify that DoubleFloat and arbitrary precision retain the input information through the output. Source scalar callbacks and their diagnostics use those same carriers.

Batching is an execution contract

Points, helicities and coefficient components have explicit ordered layouts. Native buffers retain original point identities through compaction, tail batches, failure masks and result scatter. Changing sizes, empty batches and repeated calls are covered. Headline timings use Python to coordinate native batch plans. The shared Rust coordinator now executes complete validated QCD plans in one call, including source-model gluon controls checked in all three carriers.

The coefficient scheduler uses bounded row tiles and reusable scratch storage. Its present 16-row default is a measured development choice, not a universal optimum. Dependency evaluators own their arithmetic workspaces. Binary64 can use SymJIT; high-precision paths remain native batches without an unsupported SIMD claim. Public standalone SDK integration is a separate unfinished layer.

Reducer and master interfaces

NinjaRS receives the polynomial layout, routed denominator data, exact kinematic invariants and carrier-appropriate inputs. The adapter retains weighted rank: μ^2 counts as two powers of loop momentum. Separate explicit-epsilon sectors reduce independently and combine by Laurent multiplication. A newer dense-μ numerator adapter is available as an additive internal option; the headline timings specify the actual `fast` engine they use.

The staged generic-any-n consumer (`e4c16a7`) removes the old six-propagator cap. A seven-propagator rank-eight numerator cancelling to a triangle master passes generic double/DD/AP80 and binary64 fast-tensor evaluation; broader physical multiplicity remains a separate gate. Rank above $n + 1$ still rejects.

`Runtime.load` and CLI/TOML evaluation use checked schema-v3/PACBIN artifacts and the Rust coordinator. Source gluon controls pass double/DD/AP70, independent Born/double-pole checks and native lifecycle tests. Load-time arithmetic precedes parameter binding: the CLI accepts numeric `aS=1e400` in AP70. Decimal JSON inputs, per-point scale files and all order/pole axes survive text and JSON output.

The public slice requires full-colour inclusive model-rule R_2 plus UV and explicit positive scalar or per-point μ_R^2. Generation must retain exact source, completion and scheme metadata; model names confer no certification. Generation commands, resolved/correlated output, LC/FFT, rescue and SDK/UMAMI remain open. NinjaRS packaging remains an owner follow-up.

Success, accuracy and rescue are different facts

A reducer's success flag is not an accuracy certificate. Native checks reject nonfinite input, state, coefficient and counterterm values, preserve the original failed point identity, and permit an explicit partial-result contract. Optional cross-precision checks compare complete outputs and can rescue selected failed or disagreeing points at higher precision without changing R_2 strategy.

The result records requested/evaluated precision, strategy, status and available agreement evidence. Failed points cannot silently appear as zero or be declared accurate. Existing bounded rescue controls are not a universal stability policy: threshold choices, complete feature composition and adversarial tests remain open. None of the headline runtime cells enables rescue or cross-precision verification, and each is labelled accordingly.

Known numerical gate. An equal-value, differently tracked arbitrary-precision mass construction exposed a scalar-box branch discrepancy in a CMS zero-width control. This is retained as a dependency-owner accuracy issue. Extra digits alone are not evidence that a branch-sensitive result is correct.

ARCHITECTURE · RATIONAL TERMS

Two complete strategies, one observable

The model strategy combines the four-dimensional numerator contribution with the model's effective R_2 rules and UV terms. The direct strategy instead computes the required evanescent numerator remainder and retains the same UV completion. The strategy is explicit in the result; a caller cannot relabel an executed model result as direct.

Four-dimensional reduction already produces denominator-related rational terms. The presence of μ^2 in a reducer interface therefore does not, on its own, prove that numerator-origin R_2 is complete. In the target HV scheme, effective rules already include the missing spin-dimension terms. They must not also be added to the four-dimensional numerator. FDH has a different prescription.

Local evanescent state, not a whole-loop trace engine

For a certified closed vectorlike fermion cycle, the local component construction uses dependency gamma matrices with one auxiliary evanescent slash. Odd auxiliary powers survive every open transition; only the closed trace selects even parity. Independent full-D local-graph oracles check routed bubbles, triangles and boxes. The auxiliary gamma-five realization is not a general physical chiral or dimensional gamma-five prescription.

For the admitted massless Yang–Mills cycles, dependency metric and momentum rules are continued to an explicit physical/aligned/spectator state space. The remaining transverse-state multiplicity is retained symbolically in $D = 4 - 2\varepsilon$. Local cubic and quartic colour terms, closure, routing and spectator weighting are checked independently. Ghost cycles keep their native statistics and their ordinary bare contribution.

Epsilon sectors matter after integration

If the numerator is $N_0 + \varepsilon N_1 + \varepsilon^2 N_2$ and each reduced sector has finite/single/double coefficients $\left(F_j, S_j, D_j\right)$, the assembled result is

$$V_0 = F_0 + S_1 + D_2, \quad V_{-1} = S_0 + D_1, \quad V_{-2} = D_0.$$

Dropping explicit epsilon terms can therefore change the finite answer even when the mu-dependent terms are retained. Native reduction controls exercise this convolution in all three carriers. Source templates, full-graph algebra and integrated physical results have separate checks.

What is supported internally today

The complete retained full-colour $gg \to gg$ control admits the local direct kernels, includes UV completion, and agrees with model-rule components and independent reference results in all three arithmetic modes. Bounded cross-precision verification and direct-strategy rescue are also implemented. Public direct dispatch remains behind its integration gates.

Open fermion chains, general mixed-spin loops, massive gauge/Goldstone sectors, physical axial or anomaly-sensitive structures and arbitrary BSM dimensional extensions are not inferred from that success. Unsupported direct domains need compatible model-rule completion. No universal massless FDH-to-HV shift is assumed for massive EW or general models.

Current priority. Complete effective-rule R_2 plus UV, feature coverage and higher multiplicities. Further generic direct development and strategy timing are deferred; existing bounded controls and interfaces are retained. Headline timings use model rules. The historical paired gluon LC/full cells use direct completion. Scheme distinction: QCD rational rules, §2.

VALIDATION · NEXT GATES

What this snapshot establishes

The measured controls are useful working vertical slices: model import, graphs, local recurrence, full colour, native states, reduction, UV/R_2 and ordered physical output all run together. Independent reference comparisons separate bare, local UV, external-field UV and R_2 so cancellations in the complete answer cannot conceal a wrong component.

Area	Current boundary
Complete QCD	Two-to-two controls and the five-leg quark-to-three-gluon process pass all three carriers. The six-leg dilepton-plus-two-gluon source plan now passes double/DD/AP80 at exact on-shell points; timing remains pending.
Real-mass EW	Source-owned neutral-current charged-lepton controls, including bounded massive muons; complete components and scale/parameter changes.
CMS	Explicit Gmu log choice matches MadLoop components: double/DD across widths, AP80 at finite/small widths. The AP80 zero-width master gate and Recola finite-UV convention remain separate.
Correlations / mapping	Bounded directed/spin controls and exact species symmetry; general composition and mapping transport remain open.
Public delivery	Runtime.load and CLI/TOML preserve exact inputs, all orders/poles and scale files. Explicit model/scheme declarations are integrated; generic generation, resolved/ correlated transport and SDK/UMAMI remain open.
Performance	Current development timings only; distinct-point matched provider gates and required high multiplicity remain open.

Required multiplicity and acceptance

Selected $2 \to 5$ processes remain mandatory; $2 \to 6$ is a stretch goal. Matched channels include gluon jets, heavy quarks and lepton pairs plus jets. Five-final-parton gluon channels require MadLoop/Recola controls; unavailable public OpenLoops channels are not invented for comparison.

The complete γ and Z contributions to $u\bar{u} \to e^- e^+ gg$ pass double, DoubleFloat and AP80 at three exact on-shell rational points: 16 Born graphs, 528 loops, 71 families, 406 local counterterms and 96 field insertions, with normalization $\frac{1}{72}$. Complete and separate bare/UV/R_2/field components match binary64 MadLoop/Recola within 2×10^{-8} relative, with a separate Born-scaled tolerance near zero. This is not a high-precision provider comparison. Exact double-pole and coupling-scaling identities pass 3×10^{-25} in DD and 10^{-60} in AP80. Double also passes three distinct RAMBO points, with maximum full-result relative difference 6.56×10^{-13}. Rounded RAMBO inputs still expose a DD/AP on-shell classification discrepancy; input handling and rescue remain open. No timing row is added before benchmarking.

A seven-propagator algebraic anchor passes all carriers; required $2 \to 5$ physics, broader portable plans, FFT, correlation/mapping transport and model coverage remain open. Rank above the reducer's $n + 1$ domain must reject.

Independent evidence and reproducibility

`data/measurements.json` retains the samples, revisions and cold times behind the unchanged tables; `tools/loop_reference/` contains frozen points and drivers. Timing covers complete `evaluate_ordered_native` calls, excluding checks and output conversion. New delivery and six-leg tests add no timing measurements.

Providers are MadLoop 3.7.0, Recola2 2.3.0 with matched SM/Collier, and OpenLoops 2.1.5. Tables use matched-host control timings with input and rescue asymmetries declared. Their measured object is complete physical interference.

Further reading: repository `docs/development/LOOP_MODEL_IR.md`, `docs/development/LOOP_DIRECT_R2.md`, `docs/loop-flavor-symmetry.md`, and `tools/loop_reference/README.md`; OpenLoops on-the-fly reduction · Recola · four-dimensional formulation · MadLoop standalone convention.

Living document. Update source, data and PDF after each milestone. Current priority is requested feature coverage and validated two-to-four/two-to-five processes; further two-to-two optimization is deferred. Add timing rows only after physical validation. No arbitrary caps on process size or current count; report actual dependency limits explicitly.